

Simple Math Calculator

Description

Operators

This LC3-based Reverse Polish Notation (RPN) calculator will accept numbers with up to 4 digits and perform operations:

- + - add two numbers on the stack and place the result on the stack
- - - add two numbers on the stack and place the result on the stack
- * - add two numbers on the stack and place the result on the stack
- / - add two numbers on the stack and place the result on the stack
- . - print the number on the top of stack

Prompts

The calculator will show the following prompts:

- > - ready to accept a new number
- ? - error occurred, input doesn't make sense
- \$ - stack error occurred
- ! - numeric overflow or underflow

Headings

At the beginning of the program, the calculator will display a title and a short series of subheadings to describe how to use it. As an example:

```
SMC RPN calculator
Enter 0-9 or +, -, *, /, or
. to display result on TOS
```

Conditions

Input

1. Only numbers and "." will be recognized, all other characters will be ignored.
2. If only one number on top of stack (TOS) then an math operator input, an error prompt will be displayed.
3. The print operator "." will display TOS, even if it is the only number on the stack.

Output

1. See Prompts
2. Upon error prompt, reset program to the beginning.